

Samuel Bates

Forward Deployed Engineer

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SUMMARY

Forward Deployed Engineer who turns operational risk into shipped software. At DoorDash I scoped the business case for a payment-processor migration, secured the funding, stood up a 10-agent vendor pod, and retained ~94% of at-risk monthly merchant revenue under budget, then built the Snowflake tracking dashboard that ran the program. Ships production React / TypeScript / Fastify + Snowflake (a merchant mapping engine handling hundreds of thousands of records; an 18-page workforce dashboard an external operator validated within 0.9% of the legacy system) and real AI agents in production. Owns the customer outcome end to end: scope it, ship it, measure it. ex-EMT.

EXPERIENCE

DoorDash – Forward Deployed Engineer / Technical Integrations Associate, Engineering & AI

April 2025 – Present | Remote

- **Built the Snowflake/Sigma tracking dashboard and operating model for a payment-processor migration** that retained **~94% of at-risk monthly merchant revenue under budget**. Scoped the business case with four ROI-modeled funding options, stood up a 10-agent vendor pod, and designed the operating model (Salesforce Omni-Channel queues, 3-day cadence, 200 to 300 touches/day); lifted migration conversion from under 1%/week to ~8%/week at peak and cut escalations from 20+/day to a few per week.
- **Shipped a full-stack merchant-to-account mapping engine** (React + Fastify BFF + Snowflake) replacing spreadsheet-based mapping; **hundreds of thousands of merchants flowing through production** as declared package owner. Fixed real driver-level bugs: a snowflake-sdk connectAsync() / EXTERNALBROWSER auth infinite loop (resolved with promise-based connection locking) and a silent 1,000-row query cap (resolved with a split-query plus truncation banner).
- **Built a production AI agent for merchant support (76 MAU, 6,846 queries)**. Diagnosed an adoption problem (walls of text driving abandonment), redesigned the prompt architecture into an enforced 6-section output format, and cut responses ~1,500 to ~300 words (~80%) while holding accuracy; the template was reused by 2 additional agents.
- **Built an 18-page real-time workforce-management dashboard** covering 100+ queues across multiple vendors (queue health, SLA, Erlang-C staffing) on live Snowflake data, **independently validated within 0.9% of the legacy system by an external operations lead**. Lifted an upstream name-match from ~5% exact to 95%+ via fuzzy matching; published a typed component library to an internal package registry.
- **Built and ran AI-adoption training reaching 800+ leaders across DoorDash's consumer and merchant orgs** (~180 facilitated live, the rest via reusable curricula I authored) using a DEFINE to ORGANIZE to APPLY to REFINE framework, generating follow-on AI collaborations. Authored a 22-page strategic integration briefing with 12 discovery questions to align stakeholders.
- Architected integrations across 9 enterprise systems and co-maintain the team engineering monorepo. Shipped a 22-component AI-embedded merchant-support surface, 4-week prototype-to-pilot with 19 instrumented telemetry event types, later extracted to a standalone micro-frontend that survived a same-day platform pivot. **2x Merchant Services Excellence Award**.

DoorDash – Disaster Prevention Specialist

November 2023 – April 2025 | Tempe, AZ

- Led cross-functional incident response for merchant-impacting escalations across engineering, operations, and vendor teams, resolving systemic issues under time pressure; built Salesforce / Google Sheets reporting that surfaced root-cause patterns and drove process changes reducing escalation recurrence.

DoorDash – Senior Specialist, Merchant Services

September 2022 – November 2023 | Tempe, AZ

- Scaled an integration-automation tool from 1 to 23 integration providers (tens of thousands of stores), running vendor discovery sessions to map requirements and clear onboarding blockers; built the team's knowledge base and SQL reporting pipelines.

City of Phoenix Fire Department – EMT / Behavioral Health Crisis Response Team Lead

March 2020 – September 2022 | Phoenix, AZ

- Led a 12-person crisis-response team in Level 1 trauma and stroke centers: high-stakes triage, real-time decisions under pressure, and cross-agency coordination, then taught myself to ship software at DoorDash scale.
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PROJECTS

Public builder portfolio | github.com/desertcache

- Local-first desktop AI tools: Velvet (Electron + Three.js/GLSL + on-device MLX Whisper STT), Samantha (React 19 + audio-reactive Three.js), AI Radio (Ollama + F5-TTS), and a Claude Code hooks/skills/memory scaffolding template. 21 custom Claude Code skills used daily.
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EDUCATION

Western New Mexico University

B.S. Cell & Molecular Biology, Minor in Chemistry

Estrella Mountain Community College

NREMT-B (Emergency Medical Technician)

SKILLS

Languages: TypeScript, JavaScript, Python, SQL

Frontend: React 18/19, Redux, Three.js / GLSL, typed component libraries

Backend / Data: Fastify, Node, GraphQL / Apollo (+ codegen), Snowflake, Prometheus

AI: Production LLM agents, prompt architecture, RAG, MCP, agent development, evaluation

Enterprise: Salesforce Omni-Channel, Amazon Connect, enterprise search, enterprise integration, discovery, POC delivery